Physics 2110: Midterm Exam III

(Dated: November 25, 2009)

The extra-credit problem is due in class on Monday, November 30, at 9:40 am.

1. Two rigid objects, uniform solid cylinder and thin-walled hollow cylinder, each of mass M and radius R, are released from rest at the top of an incline of height h. Which of the objects has higher velocity at the end of incline and wins the downhill "race"? (10 pts.)

2. A 60.0-cm, uniform, 50.0-N shelf is supported horizontally by two vertical wires attached to the sloping ceiling (Figure 1). A very small 25.0-N tool is placed on the shelf midway between the points where the wires are attached to it. Find the tension in each wire. (10 pts.)

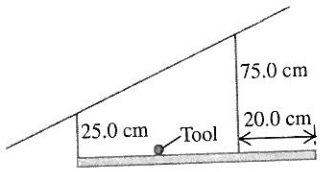


Fig. 1

3. Comets travel around the sun in elliptical orbits with large eccentricities. If a comet has speed $v_1 = 2.0 \times 10^4$ m/s when at a distance $r_1 = 2.5 \times 10^{11}$ m from the center of the sun, what is its speed when at a distance $r_2 = 5 \times 10^{10}$ m? Mass of the sun is $m_{\text{sun}} = 1.99 \times 10^{30}$ kg. (10 pts.)

4. Two springs with different force constants (stiffness) $k_1 = 5.00$ N/m and $k_2 = 3.50$ N/m, are attached to the opposite ends of a block with mass $m = 0.400$ kg on a level, frictionless surface (Figure 2). Block's motion is started by displacing it by $x = 0.050$ m to the right of the equilibrium position (chosen to be at $x = 0$). (a) Find the maximum speed the block can achieve after its release. (b) Find the angular frequency and the period of vibrations of the block. (10 pts.)

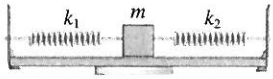
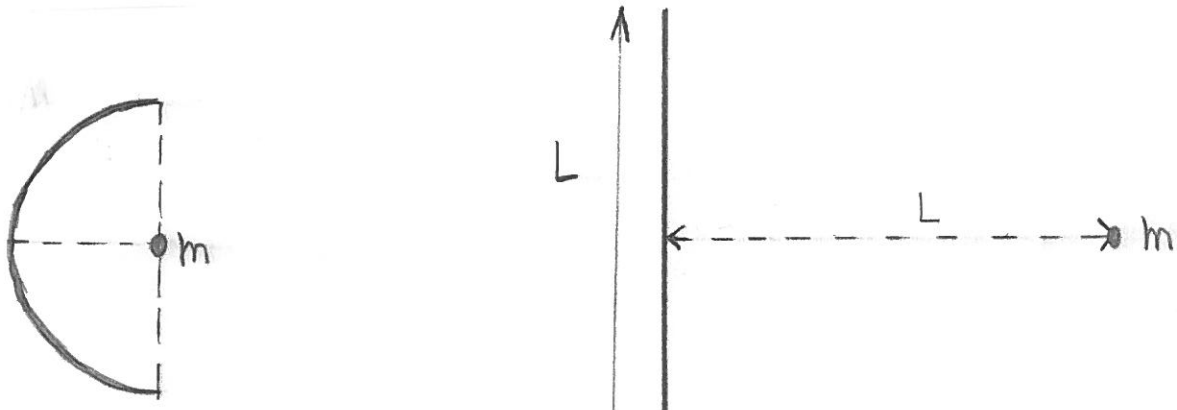


Fig. 2

5: extra credit problem. Please consult Figure 3. (a) A uniform wire with mass M and length L is bent into a semicircle. Find the amplitude and direction of the gravitational force this wire exerts on a point mass m placed at the center of curvature of the semicircle. (b) Now the same wire is straightened out and the point mass m is located at a distance L from wire's center of mass. Find the amplitude and direction of the gravitational force on the point mass m in this situation. (10 pts.)



(a) Fig. 3 (b)